The approximate approach doesn’t produce an exact solution, but there are many cases where we can settle with a sub-optimal result, or we can’t even define a perfect metric (think, for example, to retrieving the closest document to an article or the closest item to something you wanted to buy, but was out of stock). We won’t go on this path for now.

In next chapter, instead, we will delve into the latter approach, with SS-trees.

We will also defer to chapter 11 the discussion about applications of nearest neighbor search.

9.5 Summary

In this chapter we have moved from unidimensional datasets to multidimensional, spatial data. A few of the most relevant points tackled in this chapter include:

- When the number of dimensions of a dataset grows, this usually brings an exponential amount in either the complexity or memory needed.
- We need to carefully design our data structures to avoid or limit this exponential burst, but we can’t remove it altogether: when the number of dimensions is large, it’s hard, if even possible, to maintain good performance.
- K-d trees are an advanced data structure that helps performing spatial queries (nearest neighbor search and intersections with spherical or rectangular regions) more efficiently.